

Question 4.8.22:

The distance of the line

$$\mathbf{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + 5\hat{j} + \hat{k}) \quad (1)$$

from the plane

$$\mathbf{r} \cdot (\hat{i} - \hat{j} + 4\hat{k}) = 5 \quad (2)$$

is ?

Solution:

Let the line be given as:

$$\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix} \quad (3)$$

Let the plane be:

$$\mathbf{r}^T \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = 5 \quad (4)$$

Let:

$$\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} \quad (5)$$

The direction vector $\mathbf{m}$ is along the line. The normal vector $\mathbf{n}$ is normal to the plane.

Step 2: Compute $\mathbf{m}^T \mathbf{n}$

$$\mathbf{m}^T \mathbf{n} = \begin{pmatrix} 1 & 5 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = 1 - 5 + 4 = 0 \quad (6)$$

So, the line is **parallel to the plane** and does **not intersect**.

Now, we use **distance from point to plane** formula geometrically.

Since line is parallel to the plane, the perpendicular distance from **any point** on the line to the plane is the required distance.

Let us choose $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

Let d be the distance from point $\mathbf{a}$ to the plane.

Let the distance be:

$$d = \frac{|\mathbf{a}^T \mathbf{n} - 5|}{\sqrt{\mathbf{n}^T \mathbf{n}}} \quad (7)$$

Compute $\mathbf{n}^\top \mathbf{n}$:

$$\mathbf{n}^\top \mathbf{n} = \begin{pmatrix} 1 & -1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = 1 + 1 + 16 = 18 \quad (8)$$

Therefore:

$$d = \frac{|2-5|}{\sqrt{18}} = \frac{3}{\sqrt{18}} = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}} \quad (9)$$

$$\boxed{d = \frac{1}{\sqrt{2}}}$$

Line and Plane in 3D

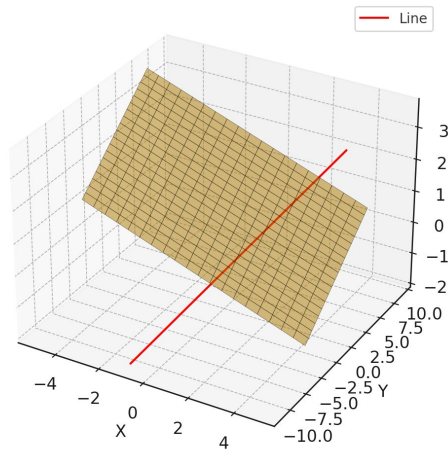


Fig. 1: Caption